Rajdeep Ghosh

IIT KHARAGPUR

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Google Scholar
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Research Interests:
Usable Security & Privacy

Education

Junior Research Fellow (PhD) *Indian Institute of Technology, Kharagpur (IIT KGP)* *June 2025 – present*

- Department of Computer Science and Engineering under the guidance of **Dr. Mainack Mondal**
- **Research Area:** Usable security— developer-centric security

MTech Degree in Computer Science and Engineering *Indian Institute of Technology, Kharagpur (IIT KGP)* *July 2023 – May 2025*

- Graduated with CGPA: 9.76/10.0 (Departmental Rank: 3)
- **Thesis Title:** Measuring vulnerabilities in NPM packages (under the guidance of **Dr. Mainack Mondal**)

BTech Degree in Computer Science and Technology *Indian Institute of Engineering Science and Technology, Shibpur (IIEST, Shibpur)* *July 2019 – May 2023*

- Graduated with CGPA: 9.29/10.0
- **Thesis Title:** Detection of Cipher Types Using Machine Learning Techniques

Publications

- [**Under Review**] Ritwik Badola, Rajdeep Ghosh, Ashita Gupta, Chester Rebeiro, Mainack Mondal. **"Anchors that Don't Lift: Understanding Supply Chain Driven Kernel Lock-In and Governance-Mediated Mitigation Strategies in SOHO Devices."** *Submitted to 35th USENIX Security Symposium USENIX Security 2026*.
- Rajdeep Ghosh, Shiladitya De, Mainack Mondal. **"I wasn't sure if this is indeed a security risk": Data-driven Understanding of Security Issue Reporting in GitHub Repositories of Open Source npm Packages** in *34th USENIX Security Symposium (USENIX Security 2025)* Usenix Security 2025 PDF arXiv version
- Abhiroop Mukherjee, Arnab Sen, Krishnendu Bera, Rajdeep Ghosh, Swarnali Mondal, Sanjana Chakravarty, Subinoy Sikdar, Malay Kuley. **Detection of Cipher Types Using Machine Learning Techniques.** in *Proceedings of Computational Intelligence in Pattern Recognition (CIPR 2023)* Conference paper

Projects

Security Issue Reporting & Resolution in Open Source Packages *IIT KGP* *2024-Present*
Conducted a large-scale study of security issue reporting and resolution in open-source ecosystems by analyzing GitHub issues from npm packages. Built ML models to detect security-related issues beyond explicitly labeled reports and identified gaps in issue classification and resolution. Ongoing work focuses on developing assistance methods for developers to help developers resolve security issues more effectively.

Kernel Lock-In and Supply Chain Security in SOHO Devices *IIT KGP* *2025-2026*
Studied Linux kernel vulnerabilities across 900+ firmware releases from 306 SOHO devices using a template-based CVE detection pipeline and supply chain tracing. Identified kernel lock-in caused by SoC SDK dependencies and explored mitigation via community, regulatory, and governance mechanisms.

ADA: Adaptive Data Assistant *IIT KGP* *2025-present*
User-centric privacy analysis system for Google My Activity logs, detecting sensitive data exposure without any data leaving the system, evaluated through a user study and interactive visualization framework.

Awards, Grants & Visits

- Invited to present lightning talk at *ACM India ARCS 2026* (IIT Hyderabad)
- Invited to present poster at Graduate Forum of *IndoML 2025* (BITS Hyderabad)
- Awarded the **Google Research Travel Grant** (of $2,750) to attend *USENIX Security 2025* (Seattle, WA)
- Received **Registration Fee waiver** for *USENIX Security 2025* (Seattle, WA)
- Secured AIR 156 in GATE CS/IT 2023 (Gate score - 787)
- Secured AIR 9967 in JEE Mains 2019 and AIR 237 in WBJEE 2019
- Recipient of JBNSTS Junior Scholar (2017-18)

Relevant Course Work

Usable Security and Privacy, Social Computing, Machine Learning, Deep learning, NLP, Foundations of LLM, Interpretable ML, Algorithms, Computer Networks, Cryptography

Skills

Human Languages: English (fluent), Bengali (native), Hindi
Computer Languages: C++, C, Python, SQL, HTML, CSS, JavaScript
Tools/Technologies: ReactJS, VueJS, TailwindCSS, Pandas, Numpy, Scikit-Learn, BS4, Latex, Git, Linux

Experience

Research Intern *Under Prof: SS Pranesh, TII, Abu Dhabi* — *Remote*
May 2022 – July 2022

- Studied working basics of fuzzer and implemented AFL fuzzer
- Contributed to understanding and development towards improving redundancy from Igor: Crash Deduplication Through Root-Cause Clustering

Web developer Intern *Menscript* — *Remote*
Sep 2020 – Feb 2021

- Created a fully responsive product display page for a product-selling website
- Redesigned the front end of the entire website using Vue.js and Bootstrap

Position of Responsibility

Teaching Assistant, B2B Training Programme on Big Data Analytics, AI-ML — NSSTA (MoSPI, Govt. of India) — *March '26*
Teaching Assistant, Data Visualisation Workshop — NSSTA (MoSPI) — *February '26*
Teaching Assistant, CS69202: Design Lab — IIT Kharagpur — *Spring '25*
Teaching Assistant, LLM Tokenisation and Embeddings session — Walmart with AI4ICPS, IIT Kharagpur — *January '26*
Teaching Assistant, Data Visualisation Workshop — NSSTA (MoSPI) — *November '24*
Teaching Assistant, CS69202: Design Lab — IIT Kharagpur — *Spring '24*
Teaching Assistant, CS69201: Computing Lab — IIT Kharagpur — *Autumn '24*

Academic Service

Artifact Evaluation Committee Member — *USENIX Security Symposium 2026*
Web Chair — *ACM ARCS 2026*